

CO2 - Assessment - 2

1. a) Data Mining :-

Data mining is the process of extracting useful pattern and knowledge from large databases.

Two Data - mining tasks : 1.) classification
2.) clustering

KDD Process :-

1.) Data cleaning

2.) Data integration

3.) Data selection

4.) Data Transformation

5.) Data mining

6.) Pattern Evaluation

7.) Knowledge presentation

- * Removes errors and missing values
- * improves data quality and mining accuracy.

b.) Data cleaning :- The process of removing incorrect, incomplete or inconsistent data.

→ Two data quality issues :- missing values; noisy data

→ Handling missing values :- * Replace with mean/median

2. a) Data Integration :- Data integration combines data from multiple sources into one data set.

Two challenges :- * Data redundancy
* Data inconsistency

* Handling redundancy :- Removes duplicate records

→ Handling inconsistency :-

standardize data formats and values.

b.) Covariance Analysis :-

covariance measures how two variables change together.

Two covariance Formulas :-

1.) Population covariance :-
$$\text{cov}(x, y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

2.) Sample covariance :-
$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

3.) a.) Stocks A and B.

covariance = +5 (positive)

* The stock prices rise together.

* They are positively correlated and not independent.

b.) Data Mining System Architecture

components :-

- * Data source
- * Database / Data warehouse
- * Data cleaning & integration
- * Database server
- * Data Mining Engine
- * Pattern Evaluation
- * Knowledge base
- * User interface (GUI)

Role :- These components work together to discover useful knowledge from data.

a.) Given

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 4, 5, 6\}$$

$$\text{Covariance} = -0.45$$

Interpretation :- Negative covariance means that variable moves in opposite directions.

b.) Data Preprocessing :-

Major steps :-

- 1.) Data cleaning
- 2.) Data integration
- 3.) Data Transformation
- 4.) Data Reduction
- 5.) Data Discretization

Pipeline :-

Raw data $\rightarrow$ cleaning $\rightarrow$ integration $\rightarrow$ Transformation
 $\downarrow$
Prepared data $\leftarrow$ Discretization $\leftarrow$ Reduction

5.) a.) Equal-Frequency Binning :- Equal frequency binning divides data into bin with an equal number of values.

For 27 values, using 3 bins:

* Bin 1 :- 13 - 22 (9 values)

* Bin 2 :- 22 - 35 (9 values)

* Bin 3 :- 35 - 70 (9 values)

b) Data Discretization

Data discretization converts continuous data into intervals (bins).

Importance :-

- + Reduces noise,
- * simplifies analysis
- * improves mining performance

